SciREX: Scientific Relation Extraction

Final Presentation

Team #3

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Contents

Overview

Research Topic

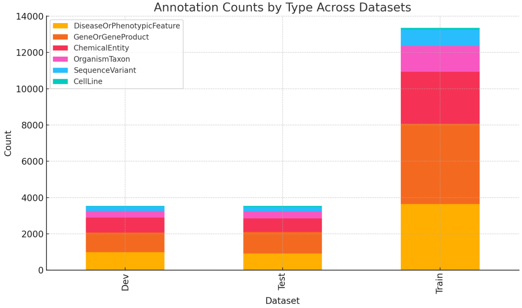
- Tackling information overload in biomedical literature as a generative information extraction task.
- Focus on automated relation extraction (RE) from scientific texts.
- Use BioRED dataset with rich biomedical entity/relation annotations as a unified benchmark.
- Goal: Scalable, automated scientific knowledge curation using LLMs.

Objectives

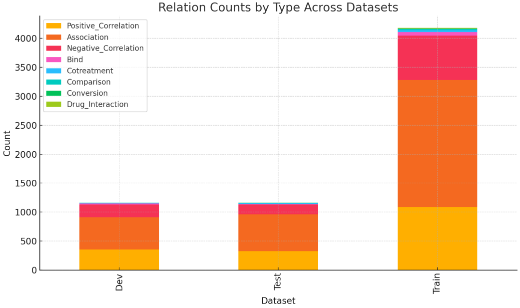
- Evaluate classification-based, LLMs instruction alignment as QA, and generative RE methods.
- Turn BioRED into a robust benchmark for comparison.
- Analyze both quantitative and qualitative performance.

Dataset Implications

- Class imbalance: rare entities and relations.
- Large training split allows diverse model training.
- As shown below, the annotation and relation distributions vary across splits.



(a) Annotation Counts



(b) Relation Counts

Figure: Entity and relation distributions across datasets.

Code: GitHub Repository

BioBERT

BioBERT for Relation Extraction

- Fine-tuned BioBERT with classification head on BioRED
- Uses sentence, entity1, entity2 $\rightarrow$ predict relation
- Fully reproducible pipeline (config-based)
- Supports full and mini datasets (via CLI flag)

Training & Evaluation Pipeline

Command-line driven experimentation using:

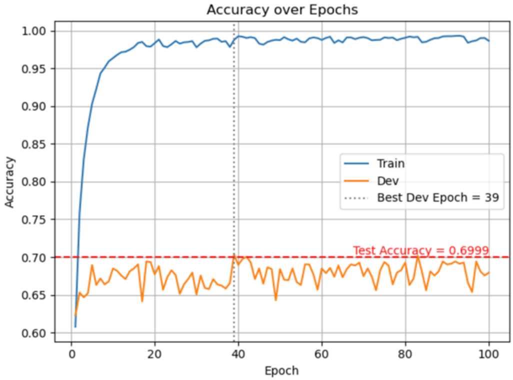
- `--dropout` : Enable dropout in classifier head.
- `--weighted-loss` : Apply class-weighted cross-entropy.
- `--weight-decay [float]` : Add L2 regularization via optimizer.
- `--label-smoothing [float]` : Soft targets to reduce overconfidence.
- `--max-seq-len [int]` : Vary input token limit (e.g., 256 $\rightarrow$ 512).

Supports: full logging, per-epoch dev reports, per-class metrics, auto plot generation.

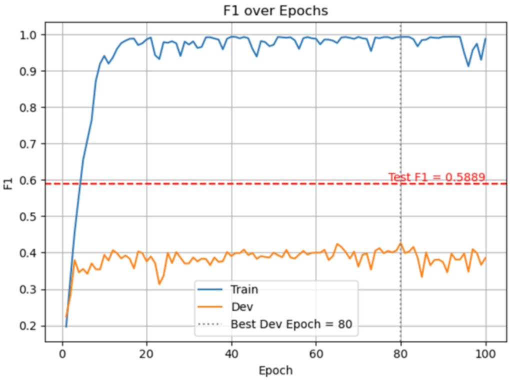
Training Performance & Evaluation

- Trained up to 100 epochs with validation-based checkpointing.
- Train F1 reaches high values early and remains stable.
- Dev F1 shows noisy improvements — best model at epoch 80 for the best run.
- Test F1 improves significantly when using best dev checkpoint.
- Overfitting visible, but mitigated through model selection.
- Class imbalance remains a bottleneck — frequent classes dominate.

Metrics over Epochs



(a) Accuracy over Epochs



(b) F1 score over Epochs

Figure: Metrics over the epochs run

Final Results & Analysis

- Total of 100+ configurations tested via CLI-driven pipeline.
- Best Test F1: **0.5889** (Baseline + 512 max tokens + 0.03 weight decay).
- **Weight decay** and **longer sequences** offered the most consistent gains.
- **Dropout** showed limited benefits when used in isolation.
- **Label smoothing** helped regularize some models (best F1: 0.556).
- Consistent logs, per-class metrics, confusion matrix, and plots enabled easy debugging.

Future Work

- Experiment with **focal loss** to handle rare class suppression.
- Run extended experiments on **higher sequence lengths** (e.g., 768+).
- Explore alternatives to BERT such as `PubMedBERT`[1], `BioMegatron`[2].
- Incorporate **data augmentation** for underrepresented relation types.
- Move to a **multi-head classification setup** to jointly predict relation + confidence.

QA4RE

Question Answering for Relation Extraction

What does QA4RE do?

- Reframe the RE task into a multiple-choice question with zero-shot instruction
- LLM selects the right relation.

Why QA4RE?

- RE makes up less than 1% of all tasks used to train LLM models.
- QA4RE aligns RE with question answering, a predominant task in instruction-tuning.
- Underrepresented task of RE $\rightarrow$ common QA format

QA4RE vs. Vanilla Prompting

Vanilla RE

Given a sentence, and two entities within the sentence, classify the relationship between the two entities based on the provided sentence. All possible relationships are listed below:
- per:city_of_birth
- per:city_of_death
- per:cities_of_residence
- no_relation

Sentence: **Wearing jeans and a white blouse, Amanda Knox of Seattle is being cross-examined by prosecutors.**
Entity 1 : **Amanda Knox**
Entity 2 : **Seattle**
Relationship: **per:city_of_birth**

QA4RE

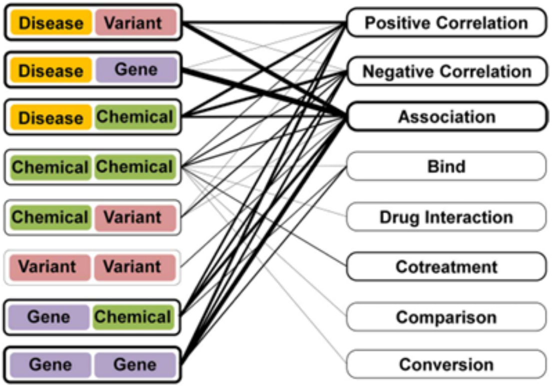
Determine which option can be inferred from the given sentence.

Sentence: **Wearing jeans and a white blouse, Amanda Knox of Seattle is being cross-examined by prosecutors.**

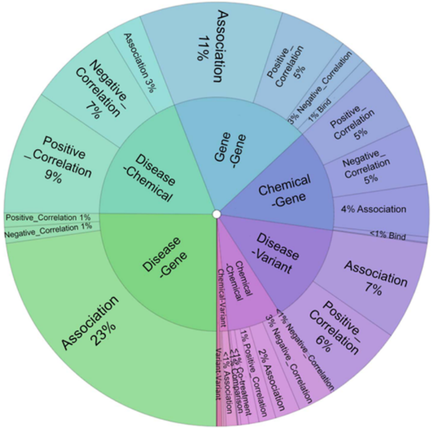
Options:
A. Amanda Knox was born in the city Seattle
B. Amanda Knox died in the city Seattle
C. Amanda Knox lives in the city Seattle
D. Amanda Knox has no known relations to Seattle

Which option can be inferred from the given sentence?
Option: **C.**

From BioRED JSON to QA4RE Prompts



(a) Relations constraints between entities in BioRED



(b) Relations distribution for entity pairs

Implemntation: Configuration and Datasets

Purpose: Evaluate multiple LLMs' performance on multiple-choice QA prompts:

Model Architecture

- **Modules:** Model loading using Hugging Face and LiteLLM
- **Seq2Seq Models**: T5-family (FLAN-T5)
- **Causal LM Models**: LLaMA, Mistral, Gemma, GPT, DeepSeek

Dataset

- **Datasets**: Train (**4497** prompts), Dev (**1284** prompts) and Test(**1123** prompts).
- **Format**: Each row is a fine-tuned prompt with its gold label (A-H)
- **Prompting**: Zero-shot vs. few-shots
- **Type Constraints**: Only valid relations (options) are offered per entity type-pair.

Implementation: Pipeline

1. Load model and tokenizer
2. For each prompt in the dataset:
   - Get prediction using model
   - Model generates a short text (e.g., "A", "B", etc.).
   - Output is post-processed to extract a valid option or "INVALID" if no valid option is detected.
   - Compare with gold label
3. Calculate accuracy and F1 scores

Results: Different Models

Only, the Test dataset is comparable with the other approaches e.g. BioBert.

Table: Model Specifications and Experimental Results on the Test and Dev Datasets

Model	Size	Architecture	Accuracy %	F1-Score%
`flan-t5-small`	60M	Seq2Seq	19	4
`flan-t5-base`	220M	Seq2Seq	20	7
`flan-t5-large`	780M	Seq2Seq	22	5
`flan-t5-xl`	3B	Seq2Seq	24	4
`Mistral-7B-Instruct-v0.3`	7B	Causal LM	21	11
`Llama-3.2-3B-Instruct`	3B	Causal LM	35	22
`Llama-3.1-8B-Instruct`	8B	Causal LM	22	15
`DeepSeek-R1-Distill-Llama-8B`	8B	Causal LM	23	15
`Gemma-3-4B-IT`	4B	Causal LM	22	2
`GPT-4o`	$\sim$ 200B	Transformer	45*	20*

*GPT-4o was tested on Dev dataset, (almost the same as the test)

Results: Top Models (zero-shots)

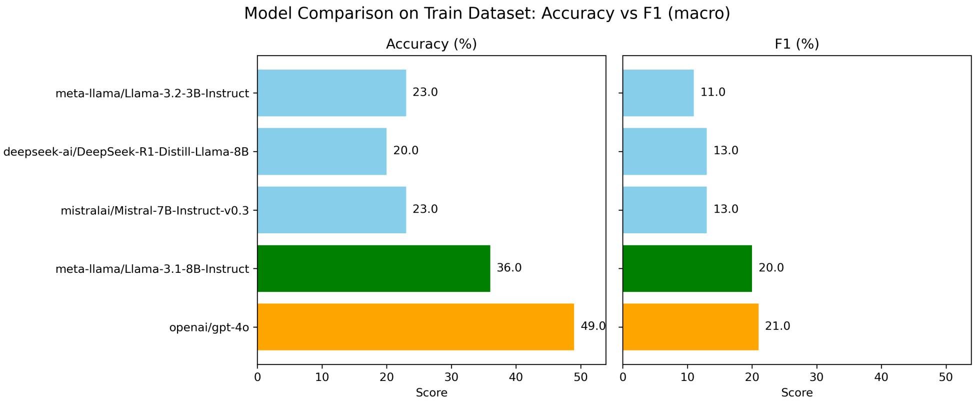


Figure: Comparison of top-performing models on the Train dataset (Accuracy and F1-score).

Results: Top Models (few-shots)

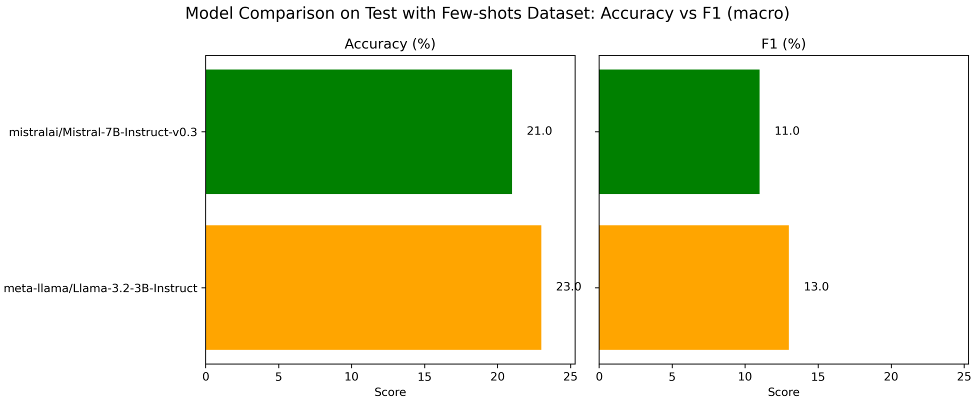


Figure: Comparison of top-performing models on the Test dataset with few-shots (Accuracy and F1-score).

Output Analysis

Observations:

- Seq2seq models (T5) have invalid predictions and **struggle** to predict biomedical RE.
- Instruction-tuned Causal LMs (e.g., Llama, Mistral, DeepSeek) achieve higher results, but less than specialized biomedical models e.g. BioBert.
- Changing the prompts used can considerably change results, showing the high **sensitivity** of wording.
- Few-shot prompting (F1 $\sim$ 13%) performed worse than zero-shot (F1 $\sim$ 20%), likely because biomedical RE requires domain-specific calibration.

Next steps improvement

- Use larger, more instruction-tuned LLMs, with domain **fine-tuning** methods such as RAG with biomedical corpora.
- Further improve prompt engineering
- Fine-tune examples for few-shot prompting with medically curated knowledge.

Scifive

Why Use SciFive?

- SciFive is trained on biomedical text (PubMed).
- It understands biomedical language better than general models.
- Based on the T5 model, well-suited for text-to-text tasks.
- Ideal for generating relation labels in natural language.

Data Preprocessing

- Extracted sentence-level context and entity mentions from BioRED.
- Paired entities to form relation candidates.
- Example: input: "a whole passage about how Aspirin treats inflammation" $\rightarrow$ output: "`Aspirin | Chemical | treats | Inflammation | Disease`"

Tokenization

- Converts text input to Token IDs and adds paddings.
- **Input Example:**

  `Aspirin | Chemical | treats | Inflammation | Disease`

- **Output:** `[259, 10, 17924, 36, 433, 36, 1964, 36, 10808, 36, 7723, 1, 0, 0, 0]`

Model Training

- Fine-tuned SciFive on formatted input-output pairs.
- Optimized model using training + validation splits.
- Saved checkpoints regularly to monitor learning progress.

Prediction

- Applied trained model to new sentences.
- Generated relation types in natural language.
- Compared outputs to held-out annotations.

Example Predictions:

- `Input: aspirin treats inflammation`
- `Prediction: . Aspirin treats inflammation.`
- `Input: BRCA1 is associated with breast cancer`
- `Prediction: susceptibility genes. BRCA1.`
- `Input: ibuprofen relieves headache`
- `Prediction: ibuprofen relieves headache, and . Oral;. Oral. Oral with migraine. Oraln or orala. Ibuprofin relieves headache headache.`

SciFive on BioRED: What Happened

- **Output drift:** lines such as `TRIPLE: E002 | Association | E007 TRIPLE OF THE PAPER...` broke the required "one triple per line" structure.
- **Hallucinated IDs:** tokens like `G011` or merged names ("tetrodotoxin LQTS") never match the gold entities, so the scorer discards them.
- **Tiny recall:** model produced only 1–2 triples per abstract (gold often has $\geq$10) $\Rightarrow$ strict recall $\approx$1–2%, F1 $\rightarrow$ nearly zero.

SciFive Takeaways

- Generative freedom without decoding guards = catastrophic scores.
- Post-processing cannot rescue malformed strings because BioRED demands exact copies.
- Lesson: SciFive needs grammar-/ID-constrained decoding or a different architecture.

REBEL

REBEL On BioRED

- Trained REBEL on the same data: outputs were cleaner, but still appended commentary (e.g. `TRIPLE CONCLUSION`) and fake IDs.
- Strict evaluation again removed most triples $\Rightarrow$ precision $\approx$0.09, recall $\approx$0.001, F1 $\approx$0.002.
- Confirms the issue is the unconstrained decoder, not SciFive in particular.

SciFive vs. REBEL

- Both models "talk" about the correct relations, yet violate the exact format BioRED requires.
- Without generation-time constraints or a copy-aware architecture, strict metrics collapse.
- Contrast: span-based models (BioLinkBERT, BiRTE) copy gold spans verbatim and dominate the leaderboard.

Generative RE: Lessons Learned

1. Add grammar/ID-constrained decoding or pointer/copy mechanisms before relying on generative IE.
2. Otherwise, stick with structured extractors for BioRED-style exact-match evaluation.
3. Document failures so future work avoids repeating the same catastrophic runs.

So, who won?

Winner

The F1-score of the models using the Test dataset:

- **BioBERT**: **0.5889**
- **QA4RE**: 0.22
- **Generative Models**: ~ 0.002

Conclusion

BioBERT

- Achieved the **highest results** of F1-score across the different methods used.
- However, it requires more time/resources to train and configure the hyperparameter

QA4RE

- Achieved **good results** of F1-score when tested on LLMs large enough with no previous training required.
- It needs careful engineering of prompts and fine-tuning.

Generative Models

- Achieved the **worst results** of F1-score as the decoder couldn't handle the format of BioRED.
- BioRed demands strict triplet formats, and generative models often produce outputs with extra words, wrong IDs, or formatting errors.

References I

[1] Yu Gu et al. "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing". In: *ACM Transactions on Computing for Healthcare* 3.1 (Oct. 2021), pp. 1–23. ISSN: 2637-8051. DOI: 10.1145/3458754. URL: http://dx.doi.org/10.1145/3458754.

[2] Hoo-Chang Shin et al. *BioMegatron: Larger Biomedical Domain Language Model*. 2020. arXiv: 2010.06060 [cs.CL]. URL: https://arxiv.org/abs/2010.06060.

[3] Jinhyuk Lee et al. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining". In: *Bioinformatics* 36.4 (Sept. 2019). Ed. by Jonathan Wren, pp. 1234–1240. ISSN: 1367-4811. DOI: 10.1093/bioinformatics/btz682. URL: http://dx.doi.org/10.1093/bioinformatics/btz682.

References II

[4] Long N. Phan et al. *SciFive: a text-to-text transformer model for biomedical literature*. 2021. arXiv: 2106.03598 [cs.CL]. URL: https://arxiv.org/abs/2106.03598.

[5] David Wadden et al. *Entity, Relation, and Event Extraction with Contextualized Span Representations*. 2019. arXiv: 1909.03546 [cs.CL]. URL: https://arxiv.org/abs/1909.03546.

[6] Ling Luo et al. "BioRED: a rich biomedical relation extraction dataset". In: *Briefings in Bioinformatics* (2022). DOI: 10.1093/bib/bbac282.

[7] Yu Su Kai Zhang Bernal Jiménez Gutiérrez. "Aligning Instruction Tasks Unlocks Large Language Models as Zero-Shot Relation Extractors". In: *Findings of ACL*. 2023.

[8] Xiao Wang et al. "InstructUIE: Multi-task Instruction Tuning for Unified Information Extraction". In: *arXiv preprint arXiv:2304.08085* (2023).

References III

[9] Yucan Guo et al. *Retrieval-Augmented Code Generation for Universal Information Extraction*. 2023. arXiv: 2311.02962 [cs.AI]. URL: https://arxiv.org/abs/2311.02962.

[10] Pere-Lluís Huguet Cabot and Roberto Navigli. "REBEL: Relation Extraction By End-to-End Language Generation". In: *Findings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: ACLicoS*. Association for Computational Linguistics, 2021, pp. 2370–2381. DOI: 10.18653/v1/2021.findings-emnlp.204. URL: https://aclanthology.org/2021.findings-emnlp.204.

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